ENERGY POSITIVE LIVING

The blueprint for India's urban future has homes that generate more power than they consume

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Photographs by ABHAY KHATRI





The most expensive building in the world is the one that keeps sending you a bill. India's residential architecture has long treated energy as someone else's problem—the utility company's, the government's and eventually the tenant's. We design the home, hand over the keys and absolve ourselves of what happens when the air conditioners switch on in May. In Delhi, they switch on in April now.

The conversation is shifting, though not fast enough. Net-zero, what a home consumes matched against what it produces, has become the fashionable ambition. In 2026, it is a modest one. The question is whether a home can give back more than it takes.

At AKDA, we have spent several years putting solar on residential roofs, as a deliberate design decision, never bolted on to satisfy a checklist. BIPV, Building-Integrated Photovoltaics, takes that same intent and scales it across the full skin of the building. In Delhi, where a typical roof is consumed by water tanks and service equipment, south- and west-facing elevations are vastly underutilised. Wrap them in solar glass and a home's power yield multiplies two to three times over. The wall that once only kept the weather out now actively pushes electricity in.

But generation is half the equation. A home that generates three times the power it needs but leaks heat through single-glazed windows in July is running two broken systems. High-performance glazing, double or triple-glazed units with low-emissivity coatings and argon-filled cavities, does the quieter work of making the envelope function. In